

By: AKH
 Rev.--- By: ---
 Chk: ---

Date May 2026
 Date ---
 Date ---

Macaulay Test

Macaulay Beam Analysis - BetaTesting

12 - Steel - Single Span -Partial UDL

Bridge, L = in
 Beam Sect = W14x61 W14x90
 A = 17.90 26.50 in²
 I = 640.00 999.00 in⁴

E = 29,000,000 lb / in²
 G = 1.740E+06 lb / in² = 0.06*29,000,000

Length Units in
 Force Units lb

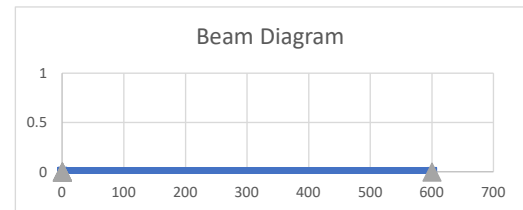
Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E	G
		in	in ²	in ⁴	lb / in ²	lb / in ²
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07	3.115E+07
	2					
	3					
	4					
	5					

Supports

	Supp. No.	X	Imposed Movement	
		in	* dY (vert)	** dr (rot)
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.000
	2	600.0	0.0	0.000
	3			
	4			
	5			

*(settlement) ** (Not Working)



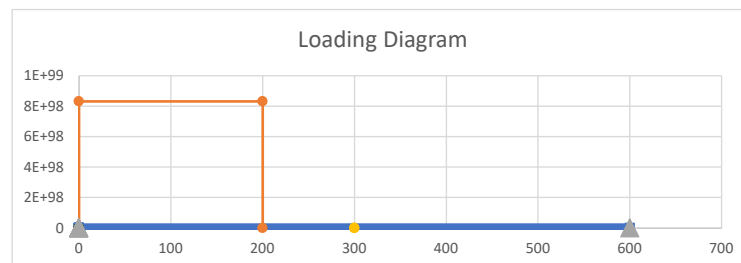
Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)
		in	in	lb / in	lb / in
DO NOT DELETE ROWS (GRAPHED)	1	0.0	200.0	-16.7	-16.7
	2	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0
	5	0.0	0.0	0.0	0.0

Load
lb / ft
200.0
0.0
0.0
0.0

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	300	0.00	0.00
	2	0	0.00	0.00
	3			
	4			
	5			



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BEAM REACTIONS

Position (X)	Enforced Disp (dy)	Enforced Rot (theta)	Vert Reac (Ry)	Internal Reac (Mz)	Note
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	2,777.8	0.0	Support reaction
600.00	0.00	0.00	555.6	0.0	Support reaction

Vert Reac
(kip)
2.778
0.556

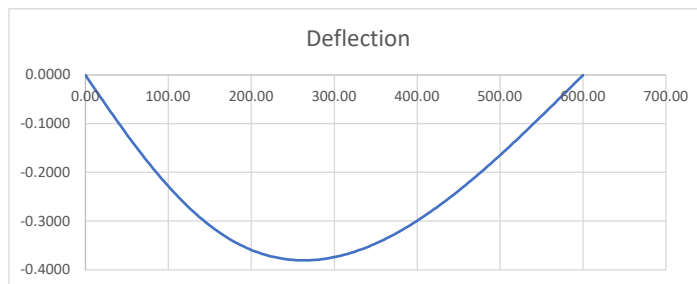
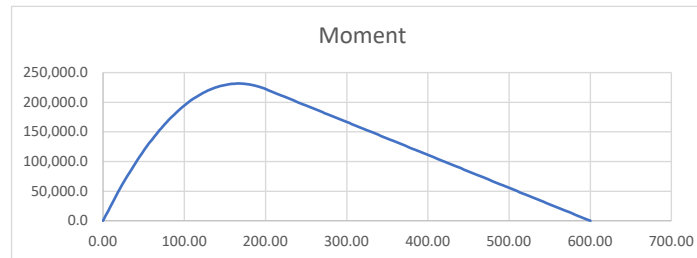
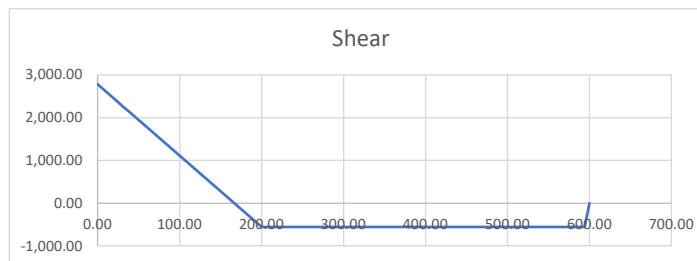
BEAM REACTIONS EQUILIBRIUM CHECK

Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-3,333.3	3,333.3	0.0	OK
Sum M @ x=0	F-L	-3.3333E+05	3.3333E+05	0.0	OK

BEAM MAX/MIN LOAD EFFECTS

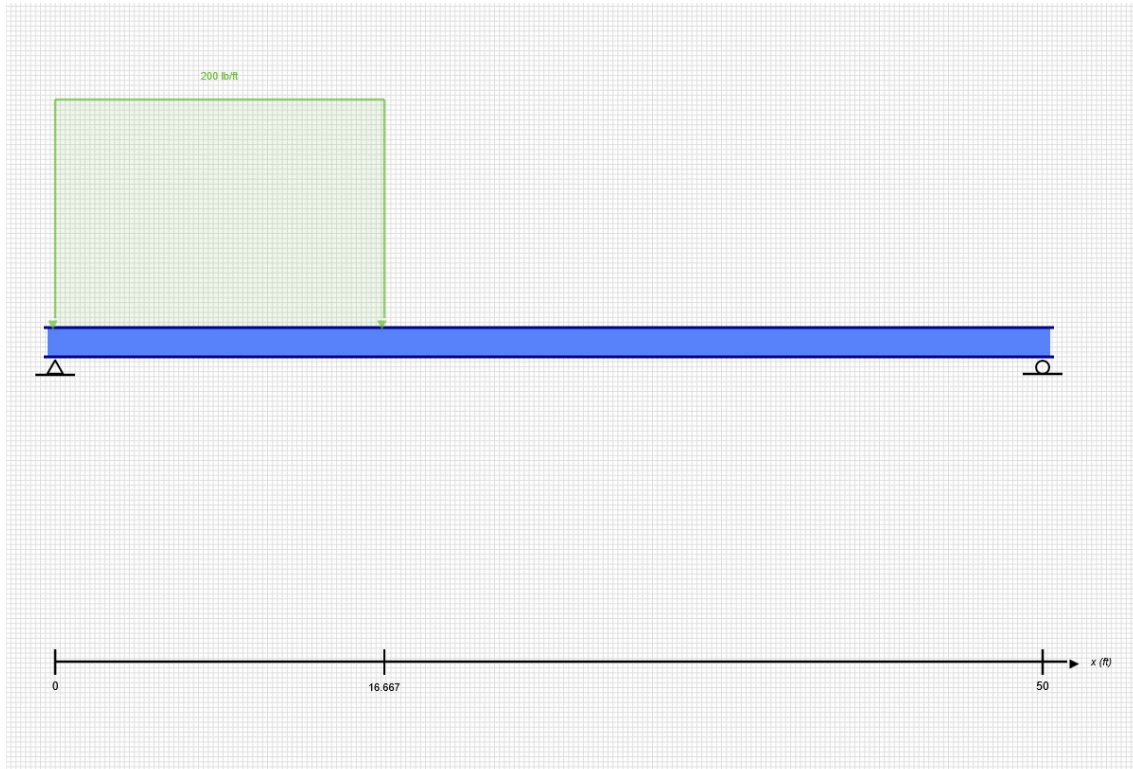
Item	Position (X)	Shear (V)	Moment (M)	Defl. (delta)
(---)	(L)	(F)	(F-L)	(L)
Max +V	0.00	2,777.8	0.0	0.0000
Max -V	200.00	-555.6	222,222.2	-0.3591
Max +M	168.00	-22.2	231,466.7	-0.3305
Max -M	0.00	2,777.8	0.0	0.0000
Max Defl	264.00	-555.6	186,666.7	-0.3806

Max/Min Value
(---)
2.78 kip
-0.56 kip
19.29 kip-ft
0.00 kip-ft
-0.381 in



SKYCIV BEAM ANALYSIS REPORT

Load Combination: DL



Software: SkyCiv Beam v3.3.2
Wed Jun 10 2026 15:27:46 GMT-0400 (Eastern Daylight Time)

Included in this Report:

- Input Summary
- Beam Section
- Free Body Diagram (FBD)
- Analysis Summary
- Analysis Results
- Bending Moment Diagram (BMD)
- Shear Force Diagram (SFD)
- Deflection Results

SKYCIV FREE TRIAL EDITION

INPUT SUMMARY

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

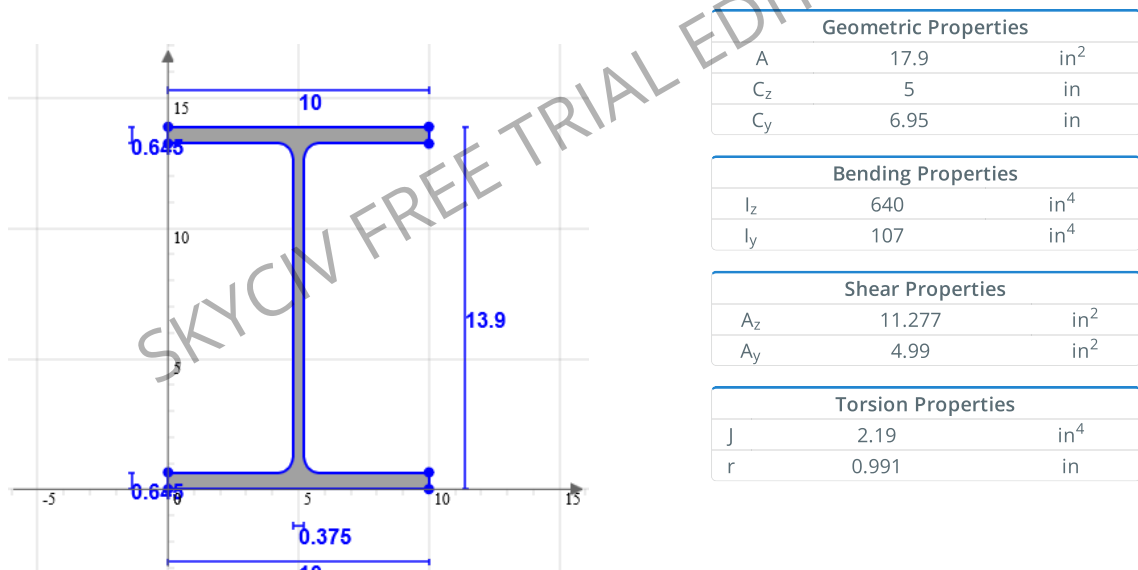
Supports

Support Type	Location
Pinned	0 ft
Roller	50 ft

Loads

Load Type	Location	Magnitude	Load Case
Distributed Load	0 ft to 16.667 ft	-200 lb to -200 lb	DL

Beam Section



Shape	Material	E (ksi)	v	ρ (lb/ft ³)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.159	L/1577
Custom Stress Limit	PASS	39 ksi	0.064	2.513 ksi
Material Yield	PASS	36 ksi	0.07	2.513 ksi
Material Strength	PASS	58 ksi	0.043	2.513 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
0	0	2777.822	0
50	0	555.578	0

Force Extremes

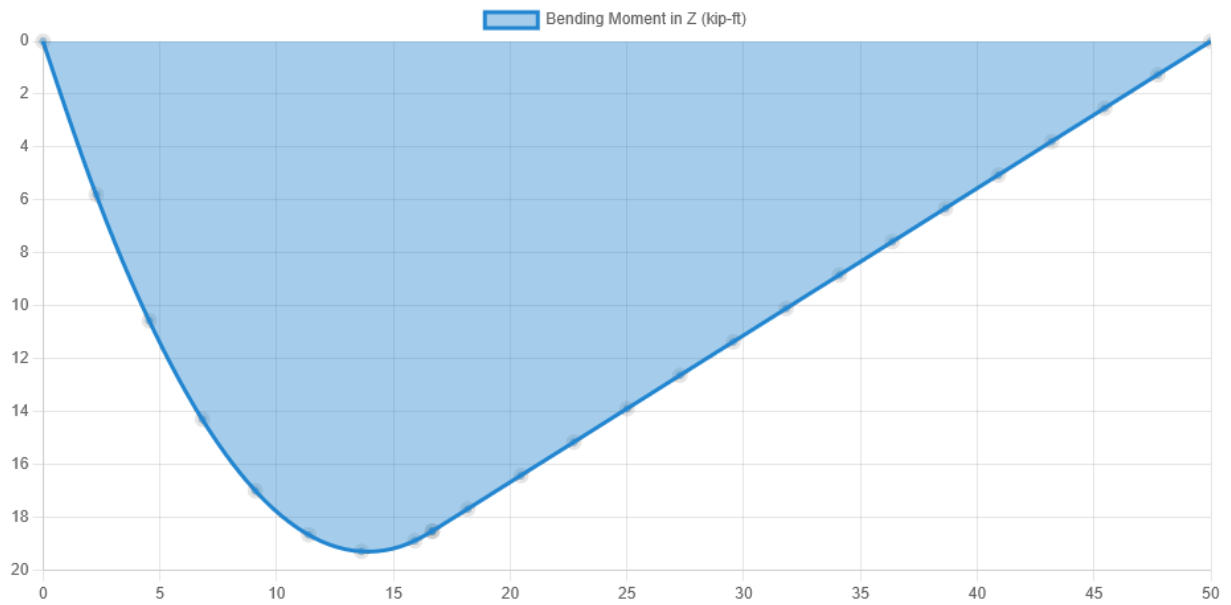
Result	Max	Min
Bending Moment	19.284 kip-ft	0 kip-ft
Shear	2777.822 lb	-555.578 lb
Displacement Y	0 in	-0.38 in

Stress Extremes

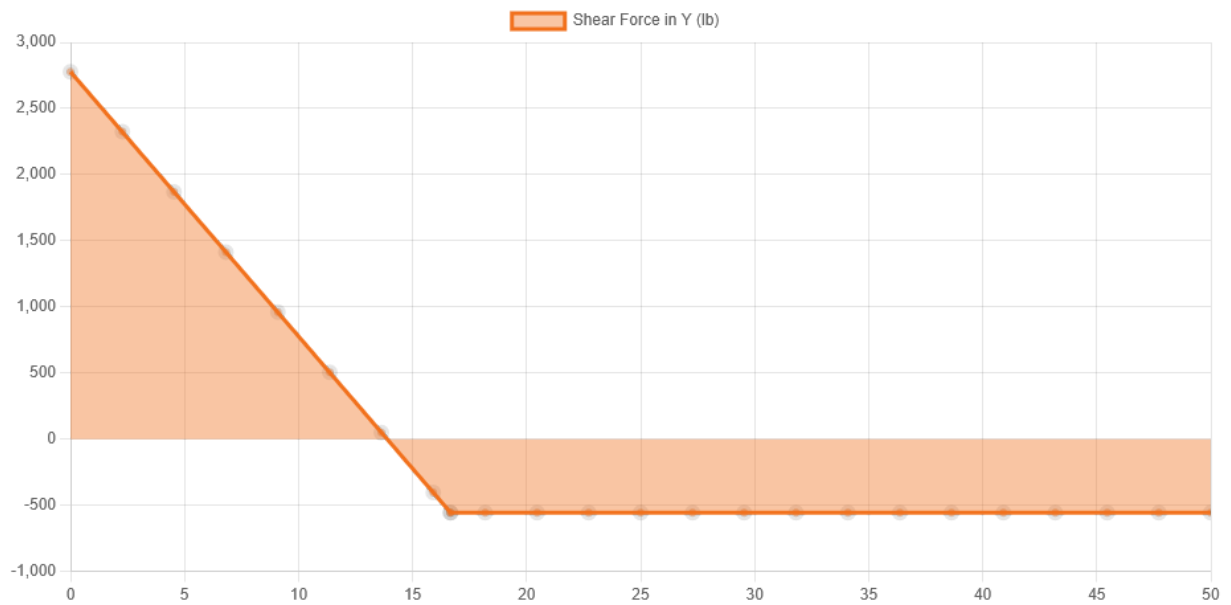
Result	Max (ksi)	Min (ksi)
Bending Stress	2.513	-2.513
Shear Stress Total	0.593	0.011
Max Combined Normal Stress	2.513	0
Min Combined Normal Stress	0	-2.513

DIAGRAMS

Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	0	22.727	50
Total Deflection (in)	0 ⓘ	0.38 ⓘ	0 ⓘ
Deflection/Span	-	L/1577	-
ⓘ The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the model settings.			